

Syllabus-Relevant Computational Geometry Q&A

Selected Exercises from Sections 33.2 and 33.3

Question 1

Show that a set of n line segments may contain $\Theta(n^2)$ intersections.

Answer 1

Yes. A set of n segments can have quadratic many intersections.

Take $n/2$ segments sloping upward and $n/2$ segments sloping downward so that every upward segment crosses every downward segment. Then the number of intersections is

$$\frac{n}{2} \cdot \frac{n}{2} = \frac{n^2}{4}.$$

So there are

$$\Omega(n^2)$$

intersections.

Also, every intersection is caused by a pair of segments, and there are at most

$$\binom{n}{2} = \frac{n(n-1)}{2} = \mathcal{O}(n^2)$$

pairs.

Therefore the number of intersections can be

$$\boxed{\Theta(n^2)}.$$

Question 2

Given two nonvertical segments a and b that are comparable at sweep position x , show how to determine in $\mathcal{O}(1)$ time whether a is above b or b is above a .

Answer 2

Compare the y -coordinates where the two segments meet the vertical sweep line $X = x$.

Let

$$a = \overline{A_1 A_2}, \quad A_1 = (x_1, y_1), \quad A_2 = (x_2, y_2),$$

and

$$b = \overline{B_1 B_2}, \quad B_1 = (u_1, v_1), \quad B_2 = (u_2, v_2),$$

where the endpoints are ordered left to right, so $x_2 > x_1$ and $u_2 > u_1$.

At sweep position x ,

$$y_a(x) = y_1 + \frac{(y_2 - y_1)(x - x_1)}{x_2 - x_1},$$

$$y_b(x) = v_1 + \frac{(v_2 - v_1)(x - u_1)}{u_2 - u_1}.$$

To avoid division, cross-multiply because both denominators are positive:

$$N_a = (y_1(x_2 - x_1) + (y_2 - y_1)(x - x_1))(u_2 - u_1),$$

$$N_b = (v_1(u_2 - u_1) + (v_2 - v_1)(x - u_1))(x_2 - x_1).$$

Then:

$$N_a > N_b \Rightarrow a \text{ is above } b,$$

$$N_b > N_a \Rightarrow b \text{ is above } a.$$

This uses only a constant number of arithmetic operations, so the running time is

$$\mathcal{O}(1).$$

Question 3

Give an $\mathcal{O}(n \log n)$ -time algorithm to determine whether an n -vertex polygon is simple.

Answer 3

A polygon is **simple** if its edges do not cross except at shared endpoints of consecutive edges. Build the n polygon edges:

$$e_i = \overline{P_i P_{i+1}},$$

where $P_{n+1} = P_1$.

Then run a sweep-line segment-intersection algorithm on these n edges.

Important Exception

Adjacent polygon edges are allowed to meet at their common endpoint. So ignore intersections between:

$$e_i \text{ and } e_{i+1},$$

and also ignore the shared endpoint intersection between e_n and e_1 .

Algorithm:

1. Form all polygon edges.
2. Use sweep line to find whether any two non-adjacent edges intersect.
3. If such an intersection exists, return “not simple”.
4. Otherwise, return “simple”.

The sweep-line algorithm takes

$$\mathcal{O}(n \log n),$$

so the polygon simplicity test also takes

$$\mathcal{O}(n \log n).$$

Question 4

A disk consists of a circle plus its interior and is represented by its center point and radius. Two disks intersect if they have any point in common. Give an $\mathcal{O}(n \log n)$ -time algorithm to determine whether any two disks in a set of n disks intersect.

Answer 4

For two disks with centers C_i, C_j and radii r_i, r_j , they intersect exactly when

$$\text{dist}(C_i, C_j) \leq r_i + r_j.$$

Equivalently, avoid square roots by checking

$$(x_i - x_j)^2 + (y_i - y_j)^2 \leq (r_i + r_j)^2.$$

To test a whole set in $\mathcal{O}(n \log n)$, use a sweep-line idea on the circle boundaries.

1. For every disk, create two events:

$$x_i - r_i \quad \text{and} \quad x_i + r_i.$$

These are the leftmost and rightmost x -positions of the disk.

2. Sweep from left to right.
3. Maintain the active disks whose x -intervals currently meet the sweep line.
4. When a disk becomes active, check whether it intersects any neighboring active disk in the vertical order.
5. When a disk leaves, remove it and check whether its former neighbors now intersect.
6. If any checked pair satisfies

$$(x_i - x_j)^2 + (y_i - y_j)^2 \leq (r_i + r_j)^2,$$

return **true**.

7. If the sweep finishes, return **false**.

There are $2n$ events. Sorting events takes $\mathcal{O}(n \log n)$, and each update/check costs $\mathcal{O}(\log n)$. Thus the total time is

$$\mathcal{O}(n \log n).$$

Question 5

Prove that in Graham Scan, points p_1 and p_m must be vertices of $\text{CH}(Q)$.

Answer 5

After choosing pivot p_0 , Graham Scan sorts the remaining points by polar angle:

$$p_1, p_2, \dots, p_m.$$

Here p_1 has the smallest polar angle and p_m has the largest polar angle from p_0 .

For p_1 , every other point lies on the same side of the directed line through $p_0 p_1$, because no point has a smaller polar angle. Therefore the segment from p_0 toward p_1 is an outer supporting direction of the point set.

So p_1 cannot be an interior point. It must lie on the hull boundary.

The same argument applies to p_m . Since p_m has the largest polar angle, all other points lie on the same side of the line through $p_0 p_m$. Hence p_m is also on the hull boundary.

Therefore:

p_1 and p_m are vertices of $\text{CH}(Q)$.

Question 6

Given a set of points Q , prove that the pair of points farthest from each other must be vertices of $\text{CH}(Q)$.

Answer 6

Let A and B be a farthest pair of points in Q . We prove both must be convex-hull vertices. Suppose A is not a hull vertex. Then A lies inside the convex hull or on a non-corner boundary part. Draw the ray starting at B and passing through A . Since the convex hull contains all points and A is not an extreme endpoint in that direction, the ray reaches a hull boundary point C beyond A .

Then

$$\text{dist}(B, C) > \text{dist}(B, A).$$

This contradicts the assumption that A and B are a farthest pair.

So A must be a hull vertex. The same argument shows that B must also be a hull vertex.

Therefore:

The farthest pair of points must be vertices of $\text{CH}(Q)$.